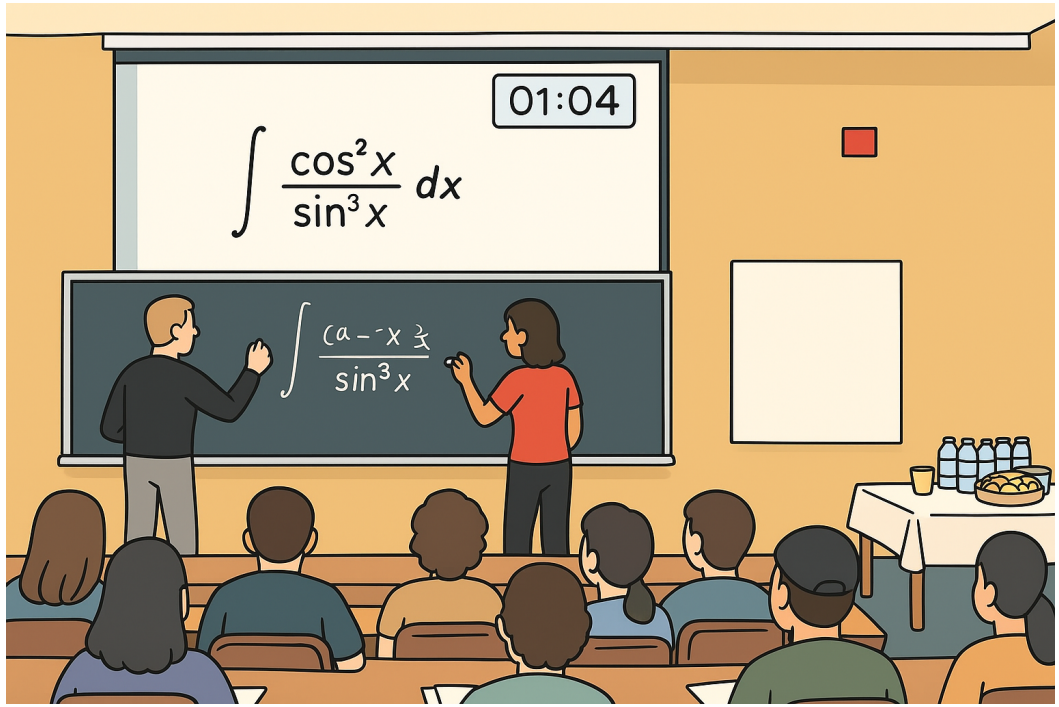
Integratión Bee Physics
Final



PRIMER TORNEO DE INTEGRALES
INTEGRATION BEE PHYSICS
CARRERA DE FISICA
UMSA

3er Lugar

Integral #13 **Integratión Bee Physic**

$$\int \frac{1}{\sin^4 x + \cos^4 x} \, \mathrm{d}x$$

Respuesta Integral #13 **Integratión Bee Physic**

$$\int \frac{1}{\sin^4 x + \cos^4 x}\,\mathrm{d}x =$$

$$\frac{1}{\sqrt{2}} \tan^{-1}\left[\frac{\tan^2 x - 1}{\sqrt{2}\tan x}\right] + C$$

Integral #81 **Integratión Bee Physic**

$$\int_0^2 x \cdot \sqrt{x^{\ln x} \sqrt[3]{x^{\ln^2 x} \sqrt[4]{x^{\ln^3 x}}}} \ldots \mathrm{d}x$$

Respuesta Integral #81

Integratión Bee Physic

$$\int_0^2 x \cdot \sqrt{x^{\ln x} \sqrt[3]{x^{\ln^2 x} \sqrt[4]{x^{\ln^3 x}}}} \ldots \mathrm{d}x =$$

$$e - \frac{1}{e}$$

FINAL

Integral #20 **Integratión Bee Physic**

$$\lim_{n\to\infty} \int_0^{\frac{\pi}{2}} x \prod_{k=1}^{n} \cos\left(\frac{x}{2^k}\right) \mathrm{d}x$$

Ayuda: Calcule

$$\sin x = 2 \sin\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right)$$

Respuesta Integral #20 **Integratión Bee Physic**

$$\lim_{n\to\infty} \int_0^{\frac{\pi}{2}} x \prod_{k=1}^{n} \cos\left(\frac{x}{2^k}\right) \mathrm{d}x = 1$$

Integral #84 **Integratión Bee Physic**

$$\int \frac{e^x}{\sqrt{\left(e^{2x}-2e^x+5\right)^3}}\mathrm{d}x$$

Respuesta Integral #84 **Integratión Bee Physic**

$$\int \frac{e^x}{\sqrt{\left(e^{2x}-2e^x+5\right)^3}}\,\mathrm{d}x =$$

$$\frac{1}{4}\frac{e^x-1}{\sqrt{e^{2x}-2e^x+5}}+C$$

Integral #100 **Integratión Bee Physic**

$$\int_0^1 \frac{\ln(x)}{1+x}\mathrm{d}x$$

Respuesta Integral #100

Integratión Bee Physic

$$\int_0^1 \frac{\ln(x)}{1+x}\mathrm{d}x = -\frac{\pi^2}{12}$$

$$= -\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^2}$$

Integral #52 Integratión Bee Physic

$$\int_0^{\infty} \frac{x}{e^x + 1}\,\mathrm{d}x$$

Ayuda: use $\dfrac{1}{1+r} = 1 - r + r^2 - r^3 + ..., \quad |r| < 1$

Respuesta Integral #52

Integratión Bee Physic

$$\int_0^{\infty} \frac{x}{e^x+1}\,\mathrm{d}x = \frac{\pi^2}{12}$$

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k^2}$$